HACKATHON SCREENING · TEAM PHOENIXAI

MIRA

AI review of Motivational Interviewing sessions



Upload a recorded counselling session. Get a transcript, an MI scorecard and timestamped coaching in about 1–2 minutes.

RN **Rahul Nakka**

DS **Deependra Shekhawat**

Azure Content Understanding · Azure OpenAI gpt-5.4 in Microsoft Foundry · LangGraph · Streamlit

Clinicians learn MI from feedback, but expert review doesn't scale

HOW FEEDBACK WORKS TODAY

Session recorded
clinician + client

→

Waits for a trained coder
MI coding needs specialist training

→

Hours of coding
every turn, against the MITI manual

→

Feedback late, or never
most sessions go unreviewed

Slow
A trained coder reviews each recording turn by turn. It takes hours per session.

Scarce
Manual review is expensive and doesn't scale, so most sessions never get feedback.

Inconsistent
Even experts disagree: two AnnoMI experts coding the same client turns agree only **κ 0.47** (0 = chance, 1 = perfect).

Needed: feedback on every session that is timely, consistent, and tied to the exact moment in the recording.

MI in one minute: draw out the client's own reasons, don't argue

Motivational Interviewing is a collaborative way of talking that strengthens a person's own motivation to change: smoking, alcohol, medication, diet, exercise.

Sal · asthma & his dog — expert-rated low quality

CLINICIAN 02:58

"Do you put Oscar higher than your baby, than your wife? That makes no sense to me."

CLIENT

"I don't think you get my situation. I've had this dog for 12 years."

0% change talk: the client argues **against** change

Mia · vaping — role-play demo

CLINICIAN 02:01

"Can we talk about what some of the pros and cons of vaping are for you?"

CLIENT

"...I think a part of me worries about getting addicted."

67% change talk: the client voices **her own** reasons to change

1 · Engaging
build trust
"I'm so glad that you came in today... this is a judgment-free zone."

2 · Focusing
agree what to discuss
"How does vaping fit into all this for you?"

3 · Evoking
draw out her reasons
"How do you see vaping influencing your performance in volleyball?"

4 · Planning
her next step
"What would you like to do moving forward in terms of vaping?"

From skills to a score: what MIRA measures

OARS: THE CLINICIAN'S FOUR CORE SKILLS

O — Open question
"What would you like to do moving forward?"

A — Affirmation
"It says a lot about who you are as a person..."

R — Reflection
simple: *"School can be so stressful."*
complex: *"You're balancing a lot on your plate."*

S — Summary
A longer reflection that pulls several points together.

WHAT THE CLIENT SAYS

change	*"I felt out of breath and super tired."*
sustain	*"Honestly, it helps me relax."*
neutral	*"I have volleyball practice basically every day."*

SCORECARD · MITI 4.2.1 THRESHOLDS · MIA'S SESSION

METRIC	MIA	BASIC · ADVANCED	POINTS
Empathy (1–5) LLM	3.5	≥ 3.5 · ≥ 4	+1
Collaboration (1–5) LLM	3.5	≥ 3.5 · ≥ 4	+1
Reflections ÷ questions PYTHON	0.50	≥ 1 · ≥ 2	0
% open questions PYTHON	100%	> 40% · > 50%	+2
% complex reflections PYTHON	67%	> 40% · > 50%	+2
% MI-adherent turns PYTHON	100%	≥ 90% · 100%	+2

8 / 12 **Basic proficiency**

0–6 Developing · 7–9 Basic · 10–12 Advanced

Client impact: change talk ratio = change ÷ (change + sustain) = 4 ÷ 6 = **67%** (no points)

How MIRA works: two AI agents, with plain code doing the maths

1 Prepare

ffmpeg · runs locally

IN
video + clip range

OUT
16 kHz mono audio; video frames stay local

2 Listen

Azure Content Understanding

IN
audio

OUT
who spoke when (ms), text, + generated fields: clinician speaker, target behaviour, summary

3 Code

Coder agent · gpt-5.4

IN
numbered turns + 22 expert-labelled examples

OUT
MITI / AnnoMI codes for every turn + a 15-word reason

4 Measure

Python · no AI

IN
turn codes

OUT
6 metrics, 0–12 score, change talk ratio

5 Supervise

Supervisor agent · gpt-5.4

IN
coded transcript + metrics + MITI rating anchors

OUT
Empathy, Collaboration, strengths, traps + better responses

6 Report

Streamlit app

OUT
score, feedback with ▶ timestamps, coded transcript, charts

AI / GENAI INSIDE

- speech-to-text + speaker separation
- LLM-generated session fields (gpt-5.4-mini)
- few-shot turn coding
- LLM supervisor with MITI anchors
- strict JSON-schema outputs
- LangGraph orchestration

TRUST BY DESIGN

- LLMs never do arithmetic
- LLMs cite turn numbers; code inserts quotes and timestamps
- Every output is typed and validated (Pydantic)

One moment, end to end: Mia at 02:29

1 Content Understanding output

```
speaker: "Speaker 1"  start: 149750 ms
text: "It says a lot about who you are as a
person… How do you see vaping influencing
your performance in volleyball…?"
ClinicianSpeaker = "Speaker 1" → clinician
```

2 Coder agent · turn T8

```
main_behaviour: question  question: open
mi_behaviour:   affirm    adherence: adherent
rationale: "Strong affirmation followed by
open question linking vaping to goals."
```

3 Coder agent · client turn T9

```
talk_type: change
"…I felt out of breath and super tired.
I just want to give my best to my team."
rationale: "Says continued vaping may
hurt performance and goals."
```

4 Python metrics

```
open questions   6 / 6  → 100% (+2)
change talk      4 / (4 + 2) → 67%
reflections ÷ questions 3 / 6 → 0.50
no LLM involved: same codes, same score
```

5 Supervisor agent output

```
strength: "Evoked discrepancy with
           her future goals"
skill:    eliciting_change_talk
turn_ids: [8, 9]   ← ids only, no quotes
```

6 Report: code inserts the evidence

▶ 02:29 Clinician: “How do you see vaping influencing your performance…?”

▶ 02:44 Client: “I felt out of breath and super tired.”

persuasion without permission

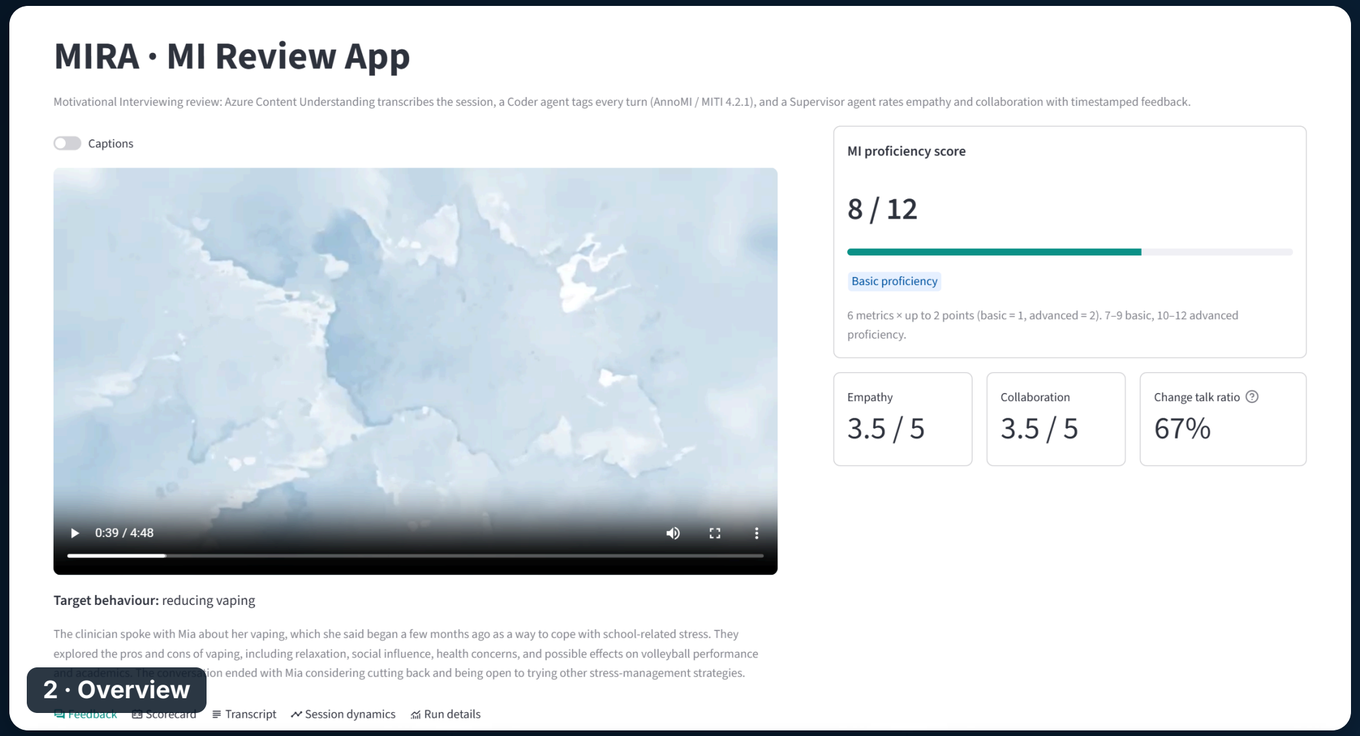
“Ask before offering strategies”

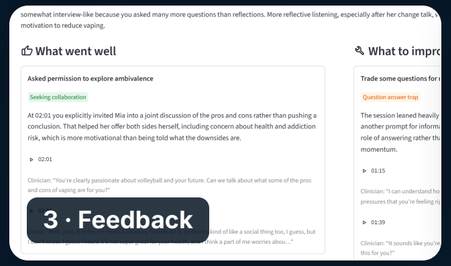
▶ 03:18

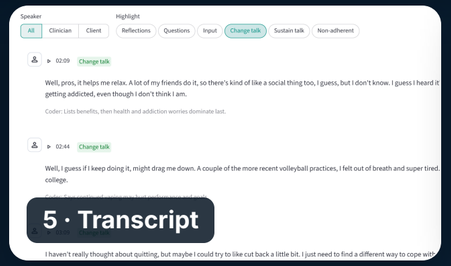
TRY INSTEAD

“I do have a couple of stress-management ideas… Would it be okay if I shared a few, and then you can tell me what, if anything, fits for you?”

Demo: Mia's session in the MIRA app









1. **Open the saved report**
   Sidebar → Saved report → Mia. No Azure call needed.
2. **Overview**
   8 / 12 Basic · Empathy 3.5 · Collaboration 3.5 · change talk 67%
3. **Feedback**
   Click ▶ 02:29: the video jumps to the moment. Read a “try instead”.
4. **Scorecard**
   R:Q 0.50 is below Basic; every metric shows AnnoMI medians for good and poor sessions.
5. **Transcript**
   Filter *Change talk*: every code comes with a one-line reason.
6. **Session dynamics**
   Change talk overtakes sustain talk.

Who it helps: coaching for clinicians, triage for supervisors

Clinician

“What should I say differently next time?”

TODAY

Score, strengths and traps with ▶ timestamps, plus a “try instead” line for each improvement, minutes after the session.

Supervisor / MI trainer

“I can't listen to every recording.”

TODAY

An evidence-linked scorecard and coded transcript to pick which sessions need a closer look. Time goes into coaching, not counting.

Programme / clinical lead

“Is our MI training working?”

INTENDED

Team-level MI-fidelity trends from the same scores. Not built yet: on the roadmap.

MI IS USED IN BEHAVIOUR-CHANGE CARE

- smoking & vaping
- alcohol & substance use
- medication adherence
- diet & exercise
- chronic conditions, e.g. asthma

GUARDRAILS

- Built on the MITI 4.2.1 standard
- Rates the clinician's technique, never the client
- A coaching aid, not a certification or clinical decision tool

AnnoMI: the expert answer key we test against

THE DATASET

133 MI demonstration transcripts (public YouTube videos)

- **110 high · 23 low** quality, rated by experts
- **Every utterance coded**: question, reflection, clinician input, client talk
- **7 transcripts coded by all 10 experts**: gives a majority vote and a human-agreement level

HOW WE USED IT



- **2** transcripts as prompt examples (60, 131)
- **27** held out for testing: 17 high / 10 low · **1,276** turns
- **104** not used

Test transcripts never appear in a prompt. A unit test guards against leakage.

1 Listen

Content Understanding transcript vs the expert transcript of the same video

2 Code

Coder labels vs expert labels, turn by turn

3 Score

0–12 score vs the expert's high / low quality rating

Check 1 · Listen: Content Understanding hears the session right

10% **word error rate**
held-out AnnoMI video 27 (3.5 min)

100% **speaker roles correct**
clinician vs client, by word

10.8% **second video** (AnnoMI 23, 12 min)
word error rate; roles 99.7% correct

WHY 10% IS BETTER THAN IT SOUNDS: ONE REAL LINE

EXPERT TRANSCRIPT ~~Mm-hmm.~~ Yeah, I ~~do I~~ definitely don't ~~wanna~~ come back here.

CONTENT UNDERSTANDING Yeah, I definitely don't want to come back here.

6 of 12 words count as errors (50% WER), yet the meaning is identical. Much of the error rate is fillers and false starts that the human transcriber kept.

WER = (substituted + deleted + inserted words) ÷ expert words. A sanity check on two videos, not a benchmark: Content Understanding also merges very short interjections such as “Mm-hmm” into neighbouring turns.

Check 2 · Code: where MIRA agrees, and where it doesn't

27 held-out transcripts · rows = expert label · columns = MIRA's label · **diagonal = agreement** · cell shading = % of the row

Reflections 644 clinician turns

expert ↓ · MIRA →	none	simple	complex	total
none	**441** 96%	13 3%	6 1%	460
simple	13 18%	**49** 66%	12 16%	74
complex	8 7%	36 33%	**66** 60%	110

- Spotting *whether* there's a reflection is easy: **96%** of “none” rows agree.
- Depth is the hard call: **36 of 110** complex reflections were coded simple.

Client talk 632 client turns

expert ↓ · MIRA →	sustain	neutral	change	total
sustain	**70** 89%	8 10%	1 1%	79
neutral	34 8%	**349** 80%	51 12%	434
change	6 5%	19 16%	**94** 79%	119

- Sustain talk is rarely missed (**89%**); change talk caught **79%** of the time.
- MIRA is generous with change talk: **51** expert-neutral turns coded as change.

Check 2 · Code: as consistent as the experts themselves

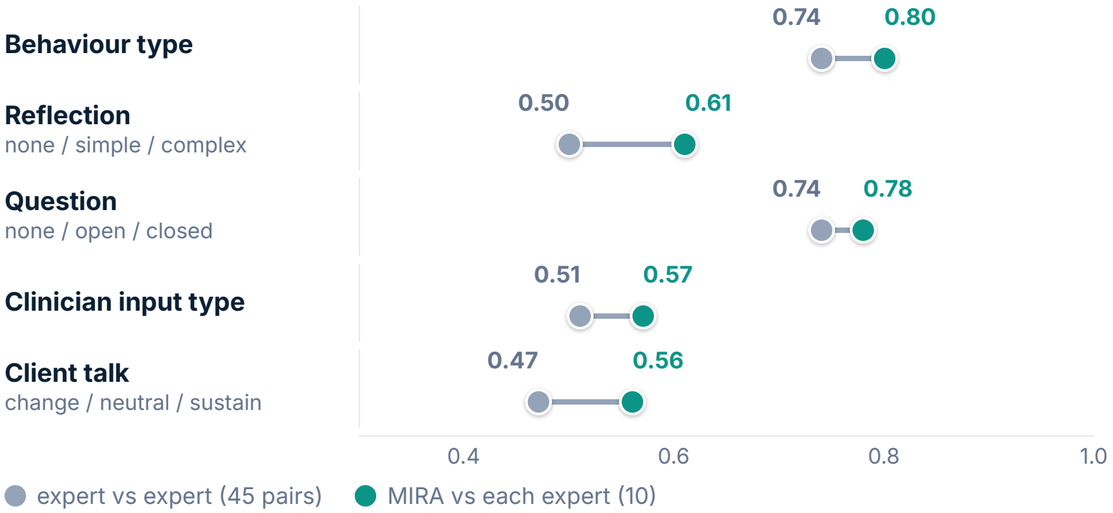
Cohen's κ = agreement after removing lucky guesses. 0 = chance · 1 = perfect. A lazy coder that calls every client turn “neutral” gets **69% accuracy but κ = 0**.

Measured on the **7 sessions coded by all 10 experts** (216 clinician · 212 client turns)

HEADLINE

MIRA as an 11th coder

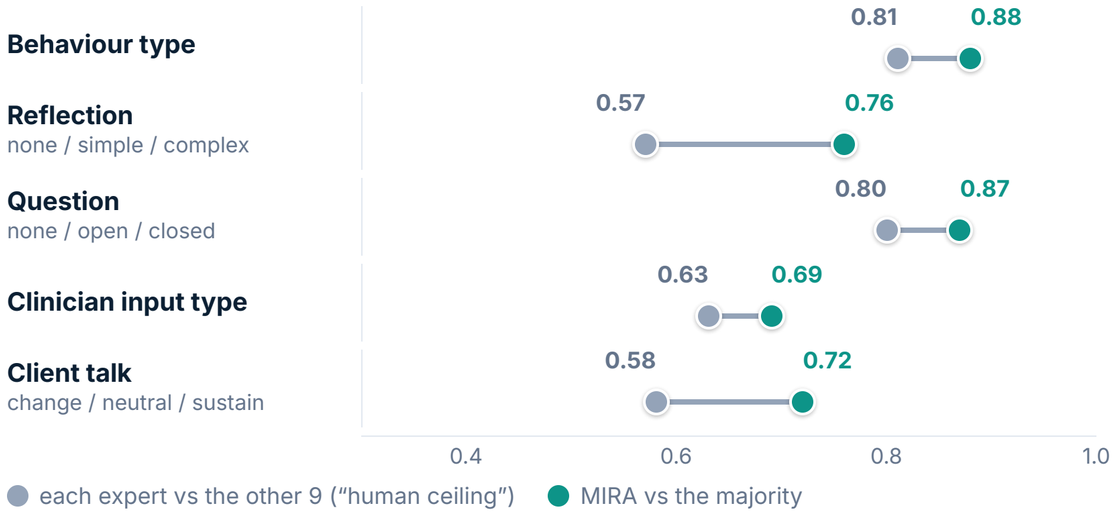
average κ against each individual expert



CONSENSUS

Against the 10-expert majority vote

a cleaner reference



Compare within a panel, not across: a single expert is noisier than a consensus. Across all 27 held-out transcripts (1,276 turns), MIRA reaches κ 0.85 on behaviour type. Model: gpt-5.4, low reasoning effort.

Check 3 · Score: MIRA's scores separate good from poor MI

MIRA proficiency score for 27 held-out sessions

each dot = one session



≤ 3 points for **all 10** expert-rated **poor** sessions

≥ 4 points for **15 of 17** expert-rated **good** sessions

164/170 good-vs-poor pairs ranked the right way (4 ties, 2 reversed)

In good sessions it flags **missed opportunities**; in poor ones, **confrontation and persuasion**.

MIRA never sees the expert's quality label (input: expert transcript text). The bands are strict: 8 of 17 good demo sessions still land in “Developing”. The score ranks sessions well, but its cut-offs are demanding.

Affordable, and kept inside our Azure

Cost per review

measured on 2 real runs

≈ $0.09
Mia · 3.1 min of audio

≈ $0.08
Sal · 3.5 min of audio



Azure list prices, Global Standard, pay-as-you-go · ~1–1.5 min per review

In the prototype today

- Only the **audio** is uploaded; video frames stay on the machine
- All AI runs in **our own Microsoft Foundry resource**: no third-party AI APIs
- **Entra ID** sign-in to Azure, or keys in a git-ignored file
- LLM calls sent with **store = false**
- Quotes and timestamps **inserted by code**, never generated
- Built on **public demo videos**: no patient data

Azure platform

- Azure OpenAI and Content Understanding are **HIPAA-eligible**, in scope of Microsoft's **BAA**
- Prompts and outputs are **not used to train** foundation models
- Data **encrypted** in transit and at rest
- Global Standard: stored data stays in our geography; inference may run in any Azure region

Before real patient data

- **Eligible ≠ compliant**: we must configure the safeguards
- Hosted app with **Entra sign-in** and role-based access
- Private endpoints, **Key Vault**, managed identity
- Encrypted storage with **retention and deletion**
- **Consent** from clinician and client

Today the app is a local Streamlit prototype with no login; recordings, transcripts and reports are kept on the local disk.

What's proven, and what's next

Proven on AnnoMI

- Codes turns at or above the human level, on **5 of 5** code families
- Ranks good above poor MI in **164 of 170** pairs
- **479 / 479** cited turns are real: no invented evidence
- About **9¢** and ~1.5 min per 3-minute review

Honest gaps

- Simple vs complex reflection is hard (experts agree only κ 0.22), so scores can shift **1–2 points** between runs
- Tested on **demonstration videos**, not real clinical sessions
- Words only: no tone, pauses or body language yet

Next

- **3-run majority vote** in the Coder for stable scores
- **Pilot** on consented real sessions, coded by MITI experts
- **Voice features**: talk time, pauses, interruptions

Thank you

Team PhoenixAI · Rahul Nakka · Deependra Shekhawat